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Course Name – Data Analytics With GenAI

Subject – SQL

Topic – SQL Basics

MySQL Workbench

SQL BASICS - PWSKILLS ASSIGNMENT

Q1. Create a New Database and Table for Employees

-- Create the database

```
create database company_db;
```

-- Use the database

```
use company_db;
```

-- Create the employees table

```
create table employees (  
employee_id INT PRIMARY KEY,  
first_name VARCHAR(50),  
last_name VARCHAR(50),  
department VARCHAR(50),  
salary INT,  
hire_date DATE);
```

Q2. Insert Data into Employees Table

```
insert into employees (employee_id, first_name, last_name, department, salary, hire_date)  
values  
(101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),  
(102, 'Riya', 'Kapoor', 'Sales', 75000, '2019-03-22'),  
(103, 'Raj', 'Mehta', 'IT', 90000, '2018-07-11'),  
(104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),  
(105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');
```

Q3. Display All Employee Records Sorted by Salary (Lowest to Highest)

```
select * from employees  
  
order by salary asc;
```

Q4. Show Employees Sorted by Department (A–Z) and Salary (High → Low)

```
select * from employees  
  
order by department asc , salary desc;
```

Q5. List All Employees in the IT Department, Ordered by Hire Date (Newest First)

```
select * from employees  
  
where department = 'IT'  
  
order by hire_date desc;
```

Q6. Create and Populate a Sales Table

-- Create the sales table

```
create table sales (  
    sale_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    amount INT,  
    sale_date DATE);
```

-- Insert data into sales table

```
insert into sales ( sale_id, customer_name, amount, sale_date)  
  
values  
  
(1, 'Aditi', 1500, '2024-08-01'),  
(2, 'Rohan', 2200, '2024-08-03'),  
(3, 'Aditi', 3500, '2024-09-05'),  
(4, 'Meena', 2700, '2024-09-15'),  
(5, 'Rohan', 4500, '2024-09-25');
```

Q7. Display All Sales Records Sorted by Amount (Highest → Lowest)

```
select * from sales  
  
order by amount desc;
```

Q8. Show All Sales Made by Customer "Aditi"

select * from sales

where customer_name = 'Aditi';

THEORY QUESTIONS

Q9. What is the Difference Between a Primary Key and a Foreign Key?

/*

PRIMARY KEY:

- Uniquely identifies each record in a table
- Cannot contain NULL values
- Each table can have only ONE primary key
- Automatically creates a unique index
- Example: employee_id in employees table

FOREIGN KEY:

- Creates a link between two tables
- References the primary key of another table
- Can contain NULL values (unless NOT NULL constraint is specified)
- A table can have multiple foreign keys
- Enforces referential integrity
- Example: If we had a 'projects' table with employee_id as foreign key,
it would reference employees.employee_id

Key Difference:

Primary Key = Unique identifier within a table

Foreign Key = Reference to a Primary Key in another table (creates relationships)

*/

Q10. What Are Constraints in SQL and Why Are They Used?

/*

CONSTRAINTS in SQL are rules applied to columns to enforce data integrity and accuracy.

COMMON CONSTRAINTS:

1. **PRIMARY KEY:** Uniquely identifies each record, no duplicates, no NULLs

2. **FOREIGN KEY:** Ensures referential integrity between tables

3. **NOT NULL:** Column cannot have NULL values

4. **UNIQUE:** All values in column must be different

5. **CHECK:** Ensures values meet a specific condition

Example: CHECK (salary > 0)

6. **DEFAULT:** Sets a default value if none is provided

Example: DEFAULT 'Active'

7. **AUTO_INCREMENT:** Automatically generates unique numbers

WHY USE CONSTRAINTS?

- Data Integrity: Prevents invalid data from entering the database
- Data Accuracy: Ensures data follows business rules
- Consistency: Maintains relationships between tables
- Error Prevention: Catches mistakes at database level
- Documentation: Makes database structure self-documenting

Example with constraints:

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE,  
    age INT CHECK (age >= 18),  
    status VARCHAR(20) DEFAULT 'Active'  
);  
*/
```